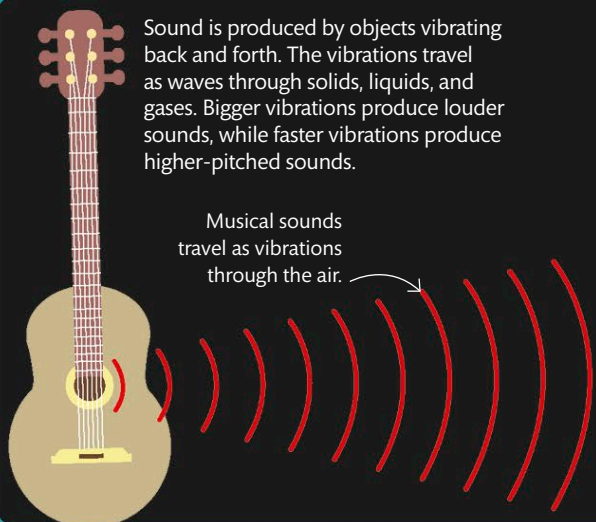


Bats and dolphins use **echolocation** (their own form of sonar) to navigate through their surroundings.

SOUND WAVES

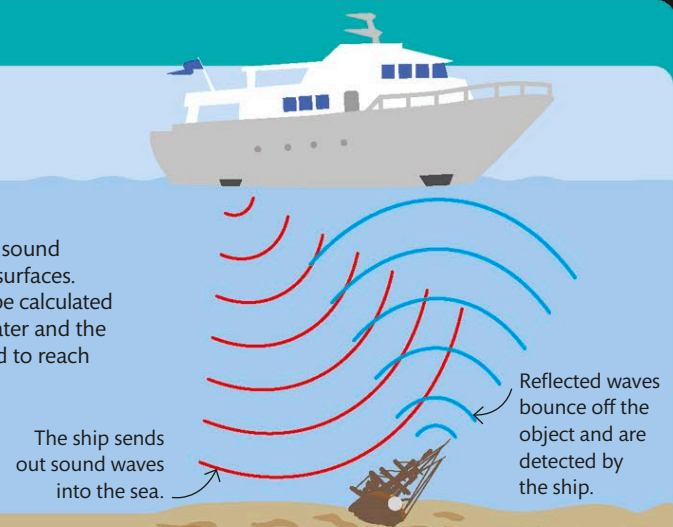


Sound is produced by objects vibrating back and forth. The vibrations travel as waves through solids, liquids, and gases. Bigger vibrations produce louder sounds, while faster vibrations produce higher-pitched sounds.

Musical sounds travel as vibrations through the air.

How sonar works

Like light waves (see page 70), sound waves are also reflected from surfaces. The distance to an object can be calculated from the speed of sound in water and the time taken for a pulse of sound to reach the object and bounce back.



The ship sends out sound waves into the sea.

Reflected waves bounce off the object and are detected by the ship.